Thomas Coen

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EDUCATION

Duke University, Pratt School of Engineering **Durham, NC**

B.S.E. Mechanical Engineering, Class of 2029 · Aerospace Engineering Certificate *GPA: 3.78*

Relevant coursework: Thermodynamics, Mechanics of Solids, Differential Equations, Engineering Design

ENGINEERING EXPERIENCE

Duke AERO (Rocketry Team) **Durham, NC**

Structures Lead *July 2026 – Present*

- Lead structural design for a 30,000 ft student researched and developed (SRAD) rocket for the 2027 International Rocket Engineering Competition (IREC) and a separate two-stage test rocket
- Troubleshoot the team's filament winder for future fiberglass and carbon tubes; completed a test wind (Oct. 2026). The stepper motor skipped and the printed gear system was too weak; the team is replacing both

Structures & Solid Propulsion Member *2025 – 2026*

- Machined a 6061 aluminum nozzle casing on a CNC lathe; hardware survived full-duration static hot fire and flight. The team placed 3rd in the 30,000 ft SRAD propulsion category at IREC 2026
- Cut prepreg carbon fiber and helped lay up rocket body tubes
- Machined canard and air brake housing components on a Viwa CNC mill
- Assembled motor hardware for static hot fire and flight
- Helped recover the rocket after flight at IREC 2026 in Midland, TX

BreakerBots, FIRST Robotics Competition Team 5104 **Pacific Grove, CA**

Co-Captain, CAD & Manufacturing Lead (2025) *2022 – 2025*

- Redesigned Leviathan's 3-stage cascade elevator (9.5 ft travel, 2× Kraken X60 motors, 1:5 reduction) with constant force spring assist, cutting extension time from 3.5 s to 1.8 s for an estimated 25% more scoring cycles (2025)
- Led a robot-wide weight reduction from 134 lb to 114 lb (15%), below the 115 lb limit. Pocketed aluminum elevator structure and replaced aluminum brackets with 3D printed carbon fiber (2025)
- Coordinated 5 subteams and 35 members to build Leviathan, the team's most technically complex robot (2025)
- Designed, CAD'd, and machined Typhoon's intake and pivot to fit inside the frame; CAM'd and machined structural parts and grew the manufacturing subteam to 15+ machinists (2024)
- Grew the manufacturing team from 2 to 8+ machinists; CAM'd and machined parts for Kraken and Cthulhu (2023)
- Machined aluminum tubing, routed the bellypan, and made polycarbonate panels for Charybdis (2022)
- Roles: machinist (2022), Manufacturing Lead (2023), Co-Captain (2024), CAD Lead (2025); Drive Coach from 2023 to 2025

Duke EGR 121 — Sunflower Seed Sheller **Durham, NC**

Mechanical Design *2026*

- Designed the rollers, drive mechanism, and funnel; did all CAD for a 4-person team
- Changed the roller shape to crack shells along the seam and reduce kernel damage
- Team produced 41 intact kernels in 3 minutes: top yield-minus-waste score among 8 section teams and 2nd among all first-year mechanical engineering teams

SKILLS

Design: SolidWorks, Onshape, Fusion 360 (CAD/CAM), AutoCAD

Manufacturing: CNC milling (Tormach 1100MX, 770M, Viwa), CNC turning, manual lathe, 3D printing, carbon fiber and fiberglass layup, filament winding, laser cutting; Titans of CNC certification series